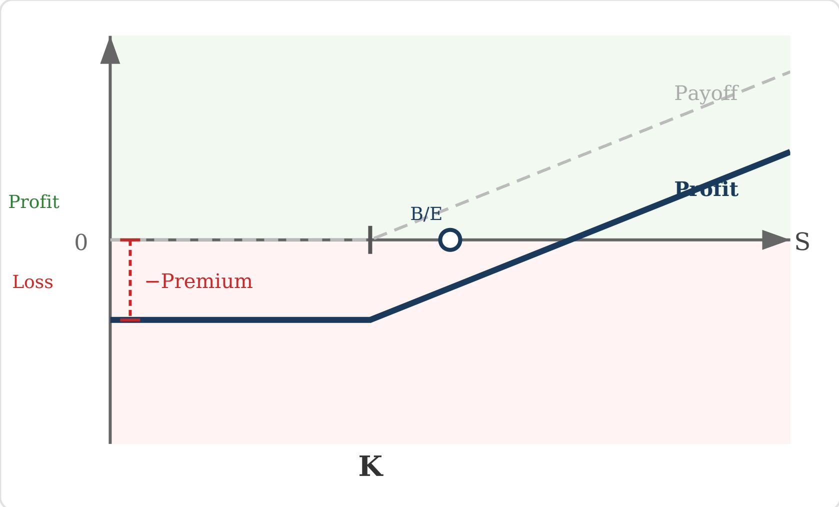
Options — Exam Reference Card

All 4 positions: identify the graph · payoff logic · corporate hedging use

Position 1 — Long Call You BUY a call option



Scenario	Outcome
S > K (in the money)	**Exercise. Profit = (S − K) − Premium**
S ≤ K (out of the money)	**Do not exercise. Loss = Premium paid**

MAX LOSS
Premium paid (known upfront)

MAX GAIN
Unlimited (S can rise infinitely)

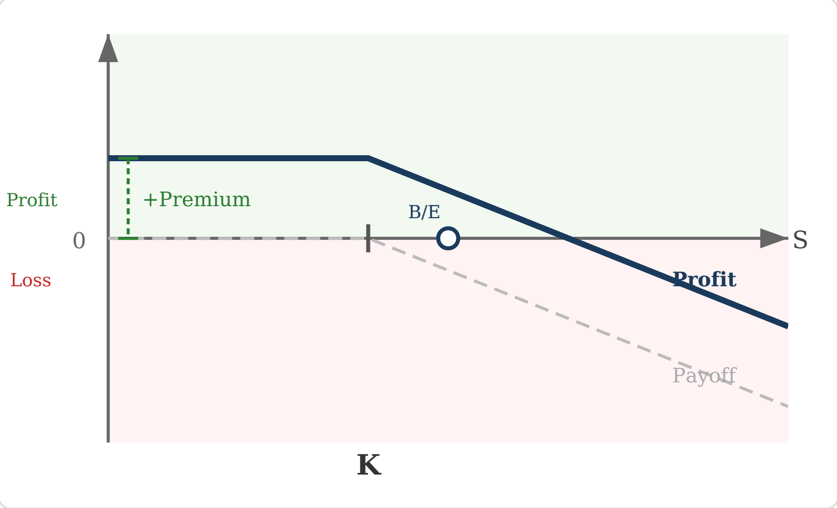
Break-even: S = K + Premium

Graph shape: Flat at −Premium left of K, then rises to the right

Corporate use: Hedges risk that an asset you need to **BUY** becomes more expensive.

> **Airline** buying jet fuel. Buys call on crude oil → caps maximum fuel cost at K + Premium regardless of how high oil goes. If prices fall, lets option expire and buys at market.

Position 2 — Short Call You SELL a call option



Scenario	Outcome
S ≤ K	**Buyer does not exercise. You keep premium. Profit = +Premium**
S > K	**Buyer exercises. You must sell at K. Loss = (S − K) − Premium**

MAX LOSS
Unlimited (stock price can rise without limit)

MAX GAIN
Premium received

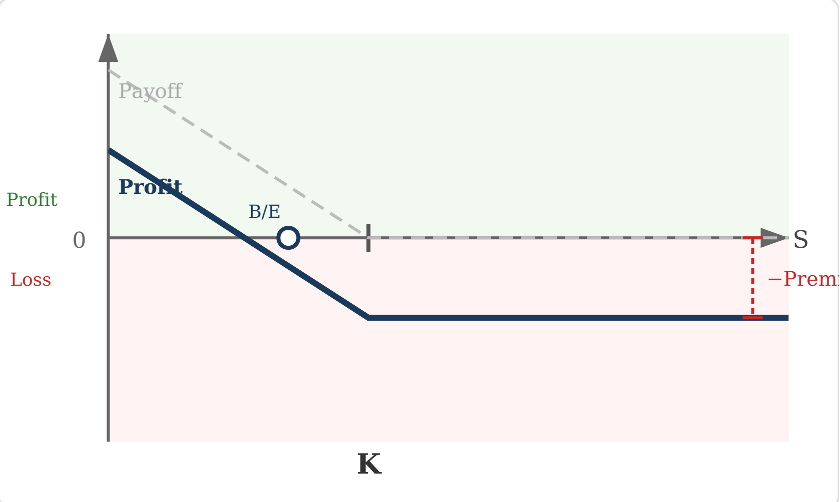
Break-even: S = K + Premium

Graph shape: Flat at +Premium left of K, then falls to the right

Corporate use: Income generation on assets **already owned** (covered call). Agree to sell at K in exchange for premium income.

> **Investment fund** holding shares sells calls on those shares. Earns premium income, willing to sell at K if price reaches it. Risk: misses upside above K.

Position 3 — Long Put
You BUY a put option



Scenario	Outcome
S < K (in the money)	**Exercise. Profit = (K − S) − Premium**
S ≥ K (out of the money)	**Do not exercise. Loss = Premium paid**

MAX LOSS
Premium paid (known upfront)

MAX GAIN
K − Premium (if stock falls to zero)

Break-even: S = K − Premium

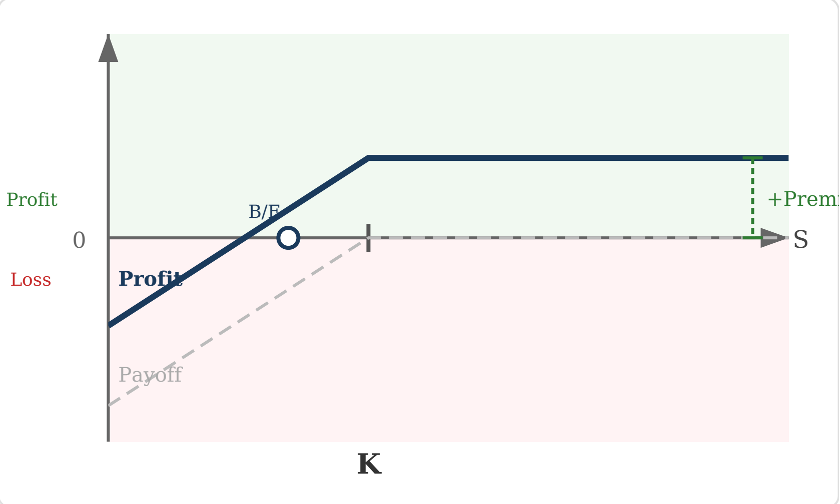
Graph shape: Rises to the left of K (higher profit as price falls), flat at −Premium right of K

Corporate use: Hedges risk that an asset you **already own** loses value. Acts as insurance.

> **Farmer** growing wheat fears harvest prices will fall. Buys put on wheat → guarantees minimum selling price K. If prices rise, lets option expire and sells at market.

Position 4 — Short Put

You SELL a put option



Scenario	Outcome
S ≥ K	**Buyer does not exercise. You keep premium. Profit = +Premium**
S < K	**Buyer exercises. You must buy at K. Loss = (K − S) − Premium**

MAX LOSS
K − Premium (if stock falls to zero)

MAX GAIN
Premium received

Break-even: S = K − Premium

Graph shape: Falls to the left of K (loss grows as price falls), flat at +Premium right of K

Corporate use: Used when a company **wants to acquire shares** at a lower price. Earns premium while waiting.

> **Company** wants to buy a stake in a supplier but thinks current price is too high. Sells puts at target price K → collects premium, and if price drops to K, acquires shares at desired price.

How to Identify the Graph in the Exam

Right side of K	Starting level	→ Position
Rising slope (↗)	Starts below zero (−Premium)	**Long Call**
Falling slope (↘)	Starts above zero (+Premium)	**Short Call**

Right side of K	Starting level	→ Position
Flat at −Premium	Starts below zero on right, rises left	**Long Put**
Flat at +Premium	Starts above zero on right, falls left	**Short Put**

Corporate Hedging — One-Line Decision Rules

You fear...	Use...	Because...
Price of something you BUY goes UP	**Long Call**	Caps maximum purchase cost
Price of something you OWN goes DOWN	**Long Put**	Floors minimum selling price
Want income on shares you hold	**Short Call**	Collect premium, agree to sell at K
Want to buy shares at a lower price	**Short Put**	Collect premium, must buy at K if exercised

Exam tip: Two observations uniquely identify any option graph: (1) Does the right side slope up or stay flat? → Call vs. Put. (2) Does the line start above or below zero? → Short vs. Long. The grey dashed line is the gross Payoff; the solid blue line is the net Profit (Payoff minus premium).